AIR UNIVERSITY, ISLAMABAD

Department of Computer Science

BS Information Technology — 2nd Semester

AYNversity

WEB TECHNOLOGY PROJECT REPORT

PROJECT TEAM		
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Course:	Web Technology
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AYNversity – Online Course Management System

Abstract

AYNversity is a web-based Course Management System developed using Angular, ASP.NET Core Web API, and MongoDB. The system provides role-based access for Admins, Teachers, and Students. Teachers can manage their courses, students can view and enroll in courses, and administrators can oversee the system. The application uses JWT authentication for secure access and MongoDB for storing user and course information.

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1. Introduction

Managing courses manually can become difficult when dealing with multiple users and courses. AYNversity was developed to provide a centralized platform where courses can be created, managed, and accessed through a web application. The project demonstrates the integration of frontend, backend, and database technologies in a full-stack application.

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2. Problem Statement

Educational platforms require an organized system to manage users and courses. Without a centralized application, managing course information and enrollments becomes inefficient. AYNversity addresses this problem by providing a web-based course management solution.

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3. Objectives

- Develop a full-stack web application.
- Implement secure user authentication.
- Provide role-based access control.
- Allow teachers to manage courses.
- Allow students to view and enroll in courses.
- Store and retrieve data from MongoDB.

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4. Project Scope

The project includes:

- User authentication and authorization.

- Course management.
- Course search and filtering.
- Course editing and deletion.
- Student enrollment tracking.
- MongoDB data storage.

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5. Technology Stack

Frontend

- Angular
- TypeScript
- HTML
- CSS

Backend

- ASP.NET Core Web API
- C#

Database

- MongoDB

Security

- JWT Authentication
- BCrypt Password Hashing

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6. System Architecture

The system consists of three layers:

1. Presentation Layer (Angular Frontend)
2. API Layer (ASP.NET Core Web API)
3. Data Layer (MongoDB)

Data flows from the Angular frontend to the ASP.NET Core API, which communicates with MongoDB for storing and retrieving information.

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7. Frontend Design

The frontend was developed using Angular.

Courses Page

Features:

- Display all courses.
- Search courses by title.
- Search courses by category.
- Search courses by instructor.
- Delete course functionality.
- Edit course navigation.

Edit Course Page

Features:

- Load course information.
- Update course details.
- Form validation.
- Loading state handling.
- Error handling.

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8. Backend Design

The backend was developed using ASP.NET Core Web API.

Main components include:

Program.cs

Responsible for:

- Controller registration.
- JWT authentication configuration.
- CORS configuration.
- JSON serialization settings.

UsersController

Handles user-related operations.

CoursesController

Handles course-related operations.

CourseService

Contains business logic for course management.

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9. Database Design

Database Name:
AYNversityDB

Database Type:
MongoDB

Collection: Users

Field	Type
_id	ObjectId
Name	String
Email	String
Password	String
Role	String

Roles

- Admin
- Teacher
- Student

Collection: Courses

Field	Type
_id	ObjectId
Title	String
Description	String
Category	String
Instructor	String
UserEmail	String
NotesUrl	String
VideoUrl	String
EnrolledUsers	String[]

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10. Authentication System

Authentication is implemented using JWT (JSON Web Tokens).

Features:

- Secure login system.
- Role-based authorization.
- Password hashing using BCrypt.
- Protected API endpoints.

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11. Seed Data

The database includes predefined records for testing purposes.

Users

- 1 Admin
- 2 Teachers
- 2 Students

Courses

Sample courses include:

- Introduction to Web Technologies
- Angular & TypeScript Masterclass
- ASP.NET Core Web API
- Database Design with MongoDB
- Full-Stack Project Workshop

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12. Implemented Features

Admin

- Manage system data.

Teacher

- Create courses.
- Edit courses.
- Delete courses.
- View their own courses.

Student

- View available courses.
- Enroll in courses.

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13. Error Handling

The application includes:

- Loading indicators.
- Request timeout handling.
- Error messages for failed requests.
- Console logging for debugging.

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14. Project Structure

backend/

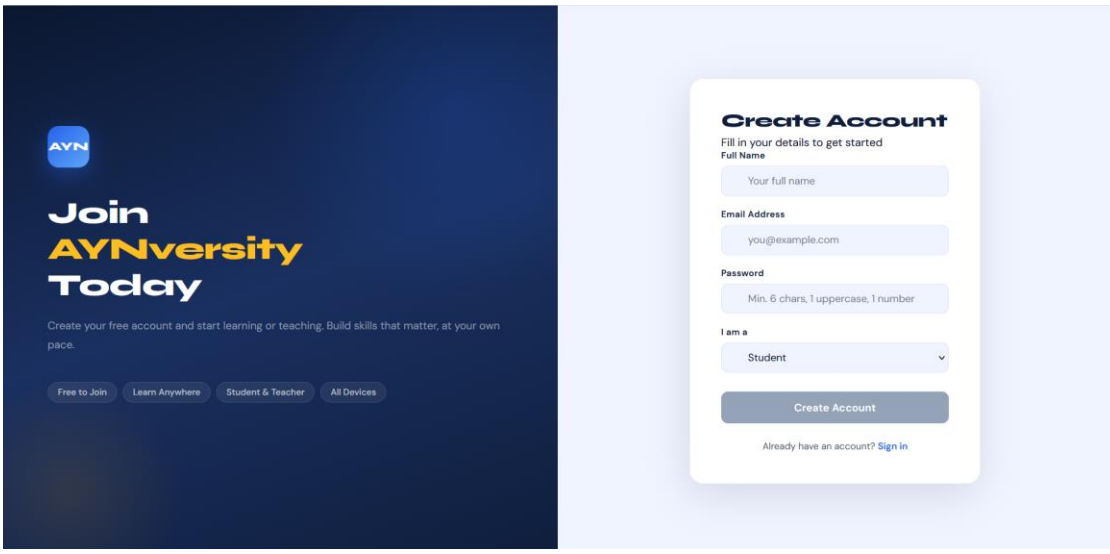
- Controllers
- Services
- Models
- Program.cs

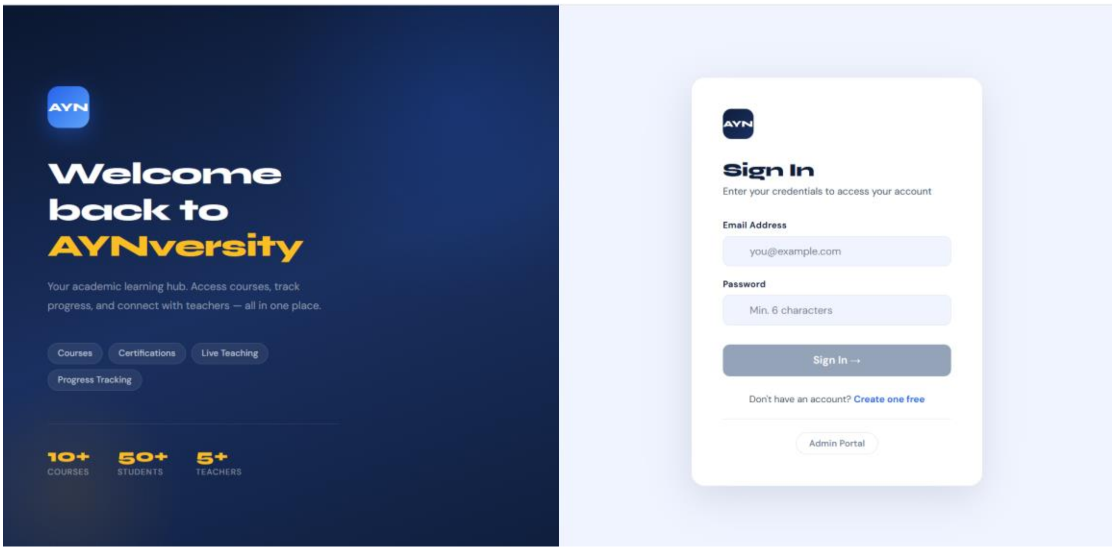
frontend/

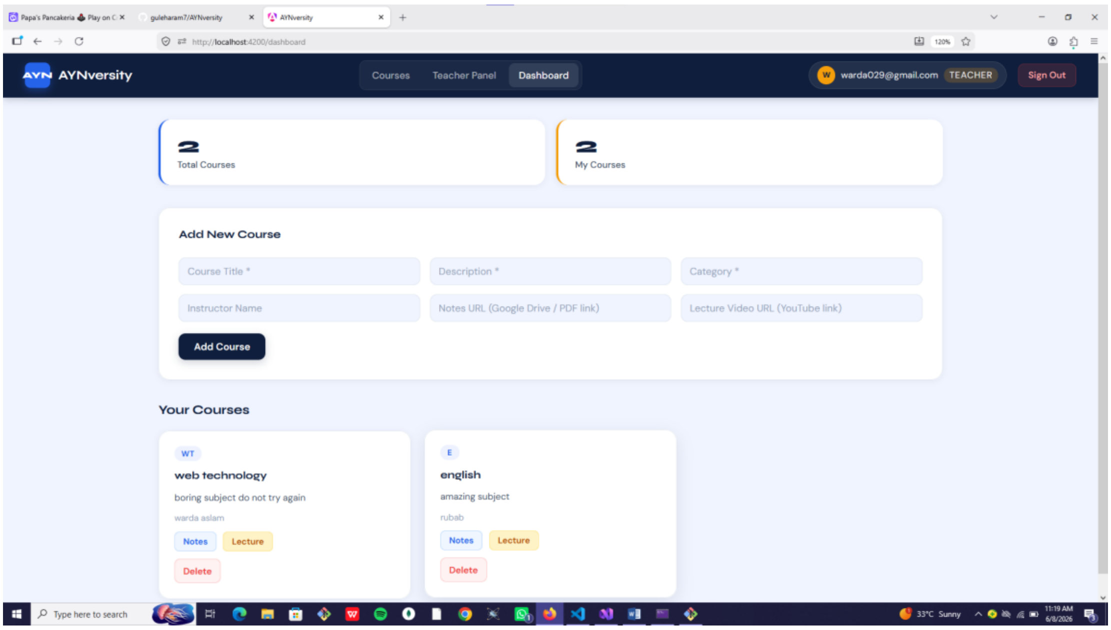
- Pages
- Components
- Services

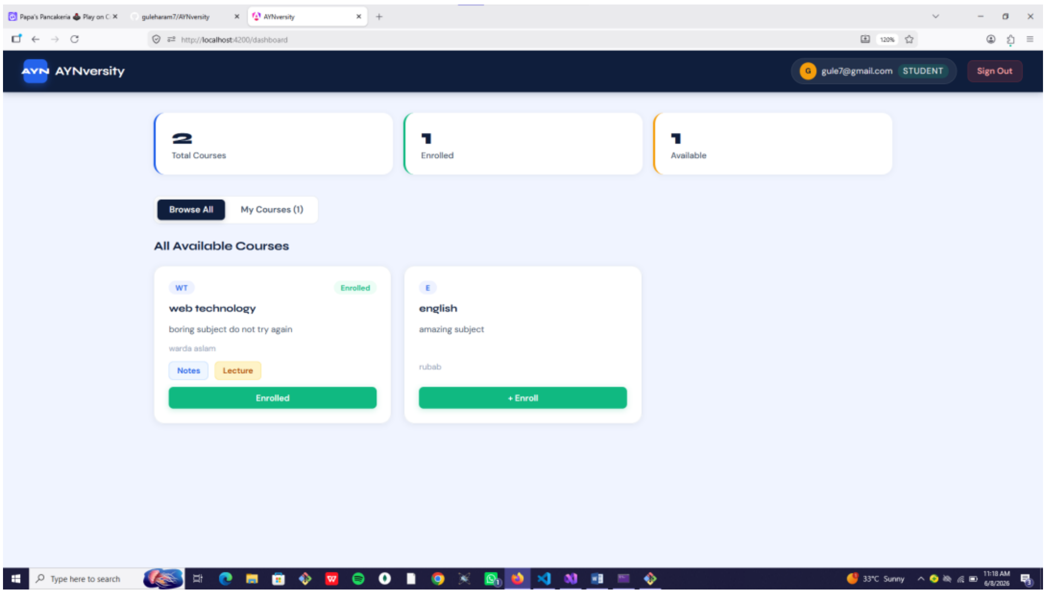
database/

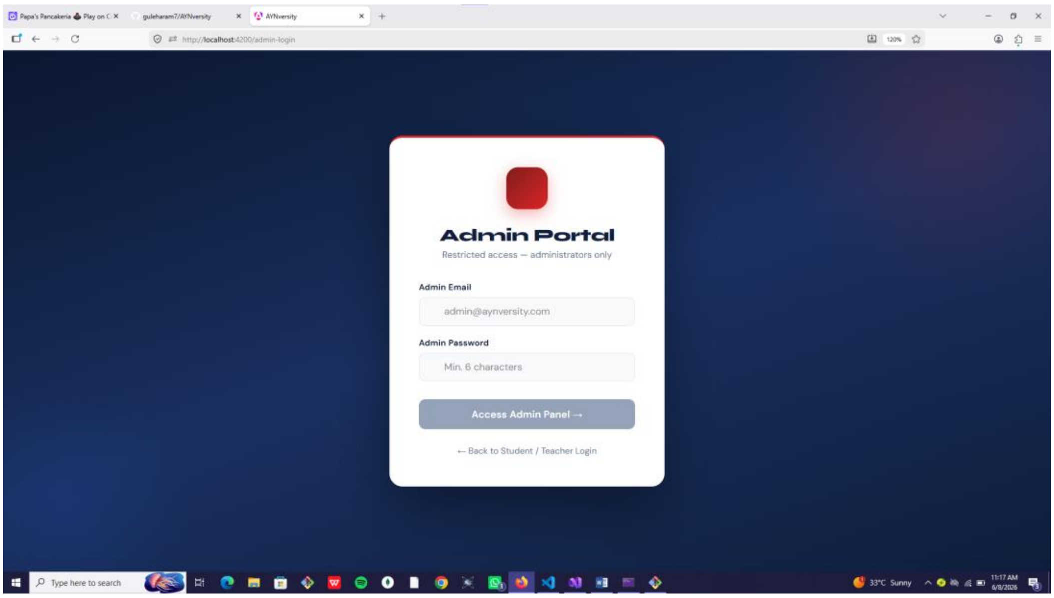
- MongoDB Seed Data

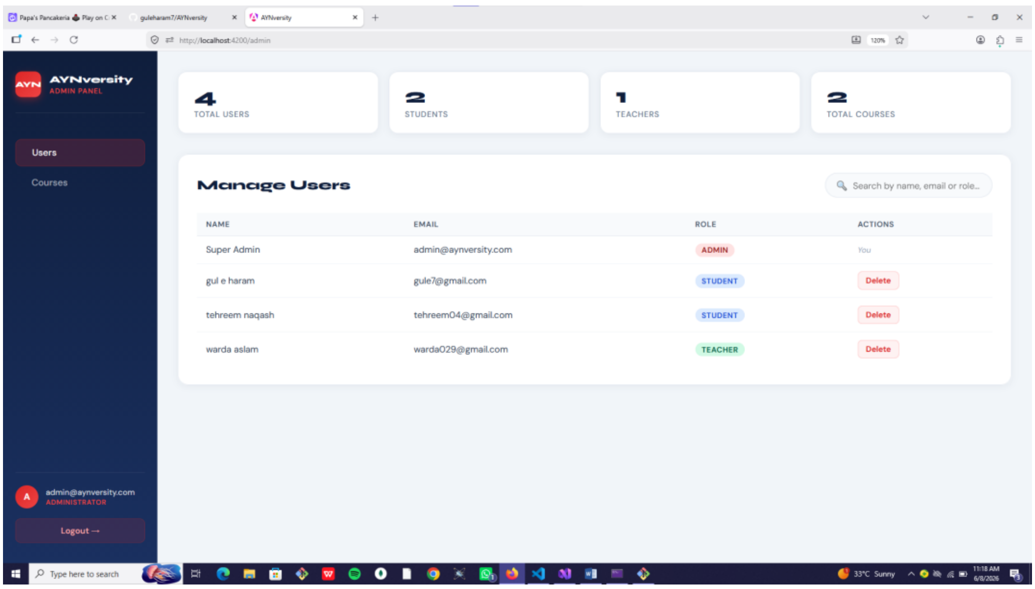

AYN
Join
AYNversity
Today
Create your free account and start learning or teaching. Build skills that matter, at your own pace.
Free to Join
Learn Anywhere
Student & Teacher
All Devices
Create Account
Fill in your details to get started
Full Name
Your full name
Email Address
you@example.com
Password
Min. 6 chars, 1 uppercase, 1 number
I am a
Student
Create Account
Already have an account? Sign in


AYN
Welcome
back to
AYNversity
Your academic learning hub. Access courses, track progress, and connect with teachers — all in one place.
Courses
Certifications
Live Teaching
Progress Tracking
10+
COURSES
50+
STUDENTS
5+
TEACHERS
AYN
Sign In
Enter your credentials to access your account
Email Address
you@example.com
Password
Min. 6 characters
Sign In →
Don't have an account? Create one free
Admin Portal


AYNversity
http://localhost:4200/dashboard
AYN AYNversity
Courses
Teacher Panel
Dashboard
warda029@gmail.com
TEACHER
Sign Out
2
Total Courses
2
My Courses
Add New Course
Course Title *
Description *
Category *
Instructor Name
Notes URL (Google Drive / PDF link)
Lecture Video URL (YouTube link)
Add Course
Your Courses
WT
web technology
boring subject do not try again
warda aslam
Notes
Lecture
Delete
E
english
amazing subject
rubab
Notes
Lecture
Delete
Type here to search
33°C Sunny


AYNversity
gule7@gmail.com
STUDENT
Sign Out
2
Total Courses
1
Enrolled
1
Available
Browse All
My Courses (1)
All Available Courses
WT
Enrolled
web technology
boring subject do not try again
warda aslam
Notes
Lecture
Enrolled
E
english
amazing subject
rubab
+ Enroll


Admin Portal
Restricted access — administrators only
Admin Email
admin@aynversity.com
Admin Password
Min. 6 characters
Access Admin Panel →
← Back to Student / Teacher Login


AYNversity
ADMIN PANEL
Users
Courses
4
TOTAL USERS
2
STUDENTS
1
TEACHERS
2
TOTAL COURSES
Manage Users
Search by name, email or role...
NAME
EMAIL
ROLE
ACTIONS
Super Admin
admin@aynversity.com
ADMIN
You
gul e haram
gule7@gmail.com
STUDENT
Delete
tehreem naqash
tehreem04@gmail.com
STUDENT
Delete
warda aslam
warda029@gmail.com
TEACHER
Delete
admin@aynversity.com
ADMINISTRATOR
Logout →

15. GitHub Repository

The project source code is maintained using GitHub.

The repository excludes:

- bin/
- obj/
- .vs/
- node_modules/
- dist/

through the .gitignore configuration.

16. Future Enhancements

Possible future improvements include:

- Course notes upload.
- Course video upload.
- Improved dashboard features.
- Enhanced user interface.

17. Conclusion

AYNversity successfully demonstrates the development of a full-stack web application using Angular, ASP.NET Core Web API, and MongoDB. The project implements authentication, role-based access control, course management, and database integration. It provides a practical solution for managing courses and users through a web-based platform.

18. References

- Angular Documentation
- ASP.NET Core Documentation
- MongoDB Documentation
- JWT Documentation

- BCrypt Documentation